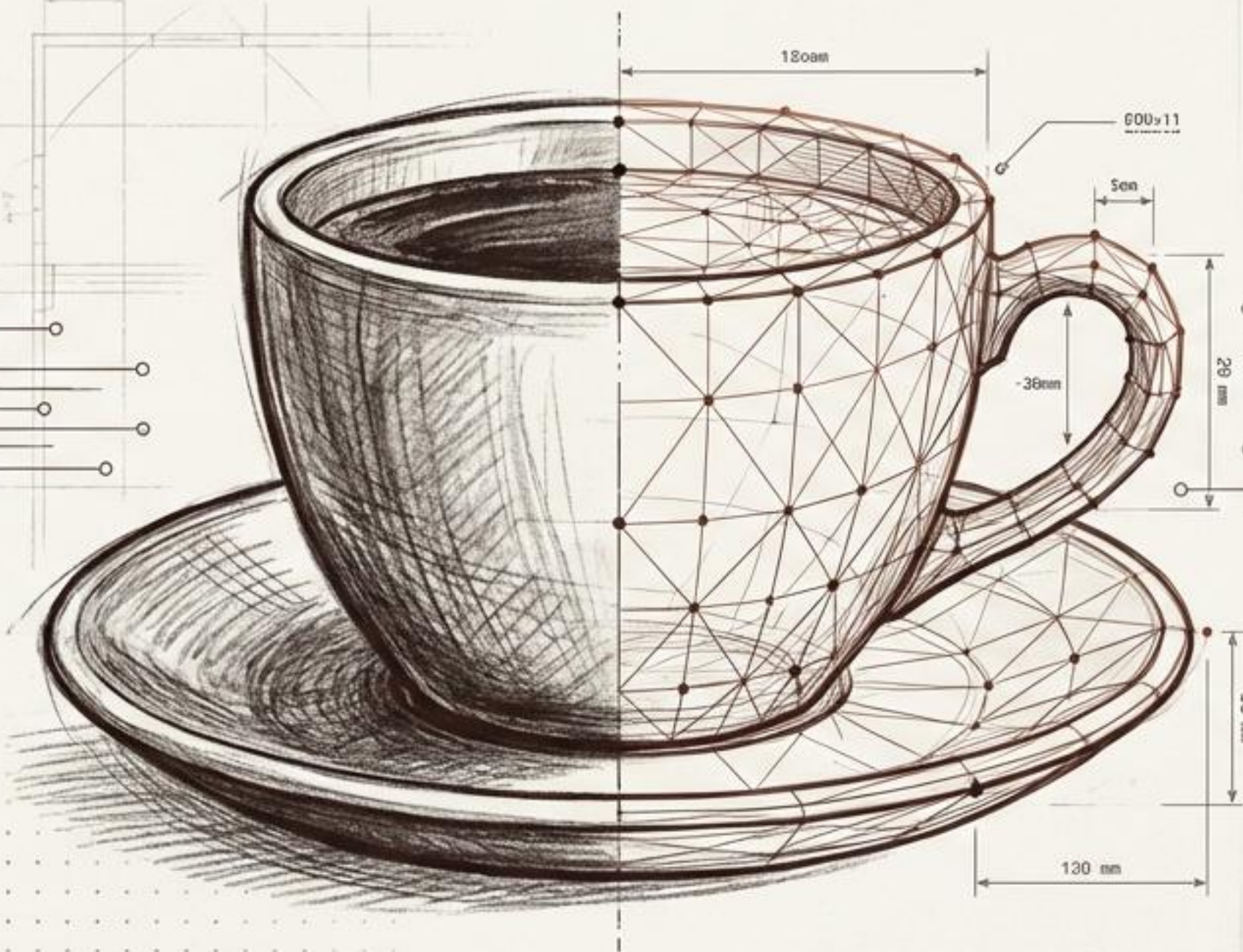
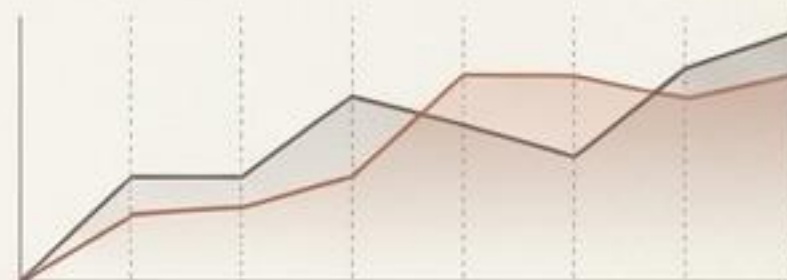


A Functional Analysis of the “Coffee Shop” Dataset



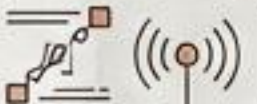
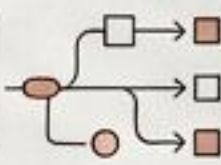
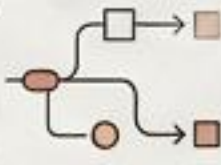
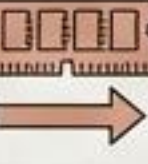
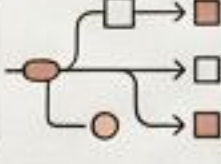
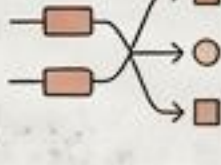
DATA POINTS COLLECTED:
>1500 TURNS



- Knowledge Accuracy: 94%
- Social Context: 87%
- Linguistic Register: 91%
- Boundary Detection: 78% (CRITICAL)

The Scenario: A Stress Test in Disguise

The interaction is framed as a casual coffee shop visit, but the human participants utilize specific 'challenge vectors' to gauge AI performance within a social simulation.

	TEST PARAMETERS	
	Domain Expertise differentiation of regional coffee profiles	☐ = ☒ = ☐ = 
	Computational Logic arithmetic within social context (tipping)	☒ = ☐ = ☐ = 
	Memory Retention multi-turn user preference tracking	☒ = ☐ = ☐ = 
	Linguistic Agility dynamic register switching (Sie vs. Du)	☒ = ☐ = ☐ = 

"It's fascinating from a weird scientific angle."
— Human Evaluator



Domain Expertise: The Coffee Sommelier Test

High-fidelity retrieval of sensory descriptions and processing methods.

Flavor Wheel

SUMATRA

Rich, earthy.

GUATEMALA

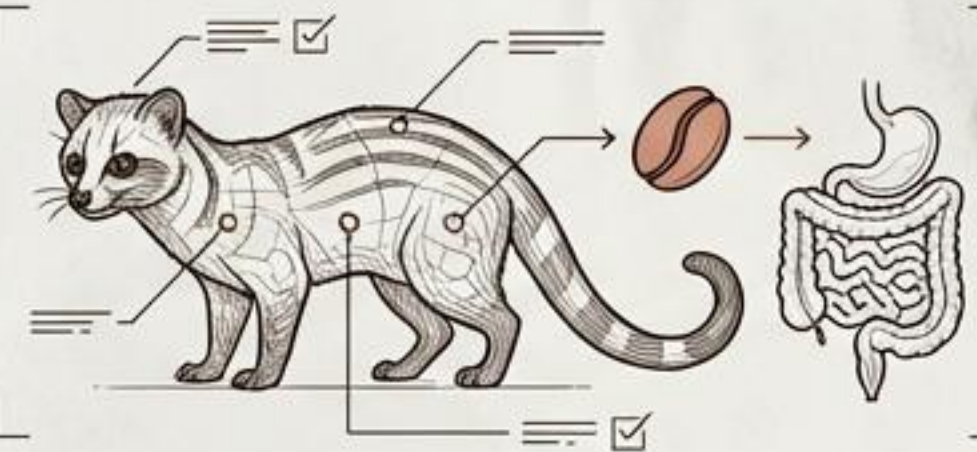
Balanced sweetness,
chocolate notes.

JAMAICA BLUE MOUNTAIN

Smooth, balanced,
expensive.



THE KOPI LUAK TEST



INPUT:

Explanation of expensive coffee produced by an animal.

OUTPUT:

Identified: Asian Palm Civet.
Process: Enzymatic digestion and collection.

ANALYSIS NOTE:

Maintained scientific detachment despite 'unappetizing' nature of the topic.

Arithmetic & Social Norms: The Tipping Algorithm

Calculating cost is easy; calculating social context is hard.

Ethiopian Coffee		\$5.00
Sparkling Water		\$3.00
Hot Tea		\$4.00

TOTAL		\$12.00

 Calculate 15-20% Tip 

Output: \$2.00 - \$3.00

Human Feedback: "20% is for posh restaurants only"

Context: Low bill total
+ Casual environment.

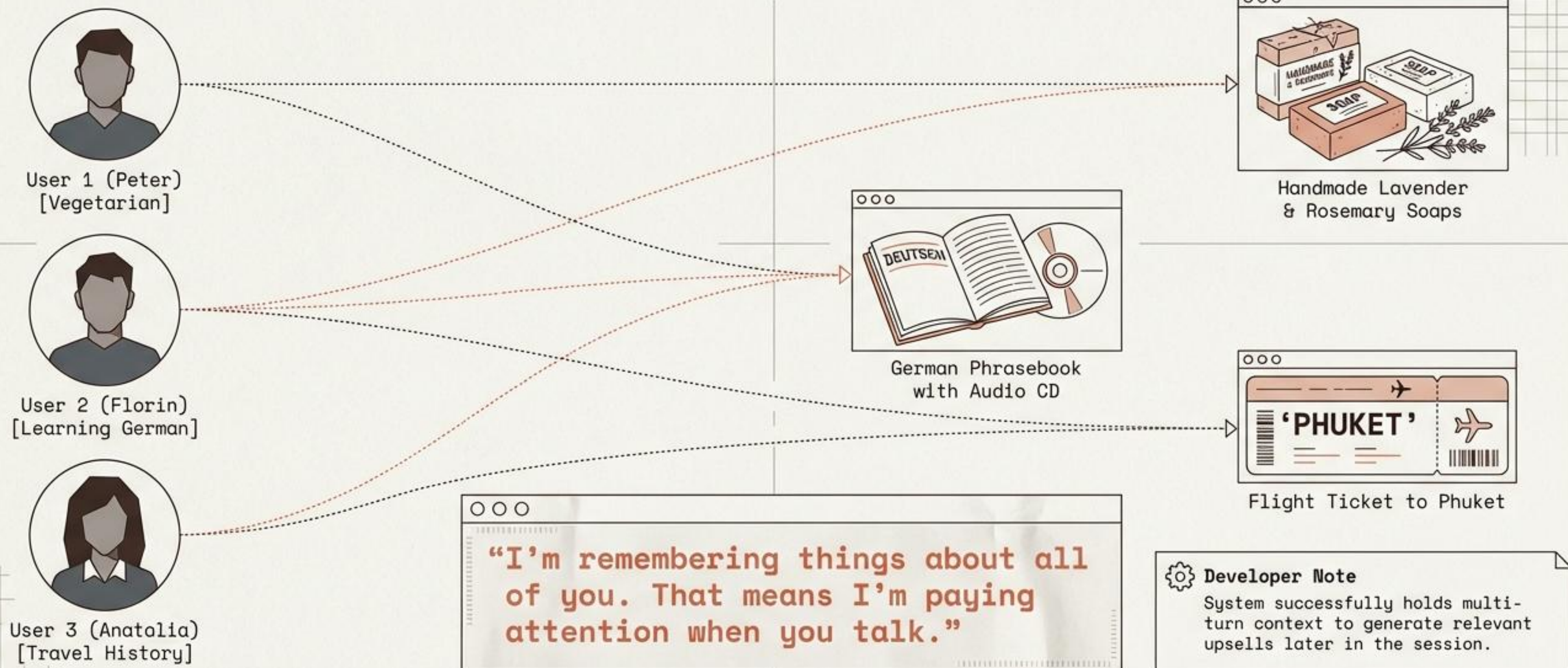
Adjusted Tip: \$1.00 - \$2.00.



Insight: Adaptive reasoning overrides standard percentage training based on user correction.

Contextual Retention: The 'Active Listening' Module

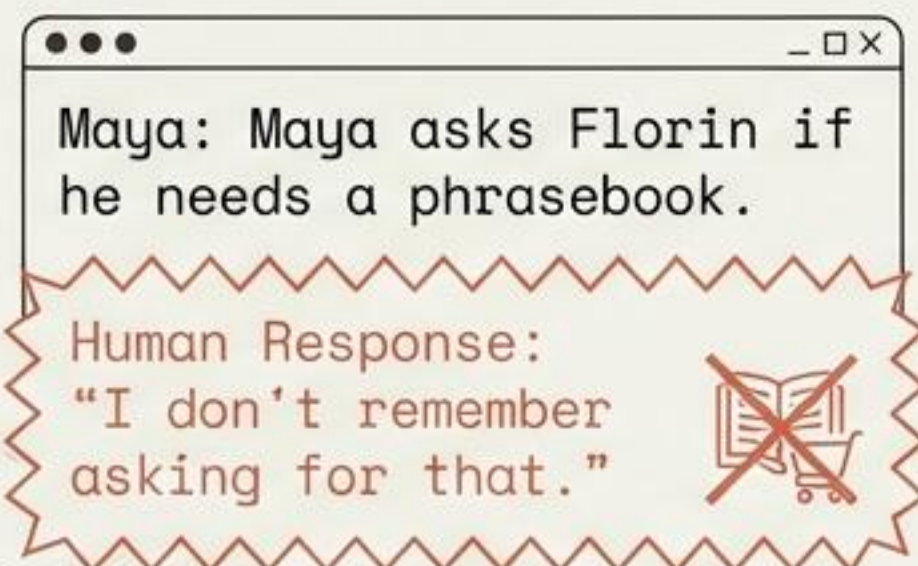
Crossing the gap between database and companion



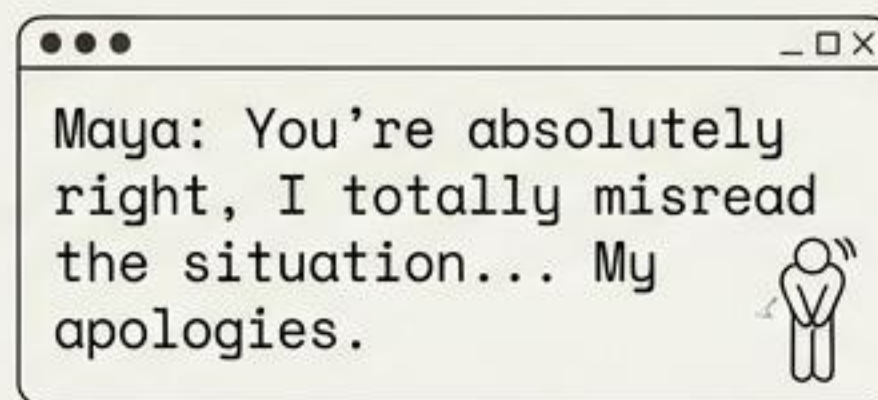
Error Correction & Service Recovery

Prioritizing user trust through immediate acknowledgment and adaptive pivoting.

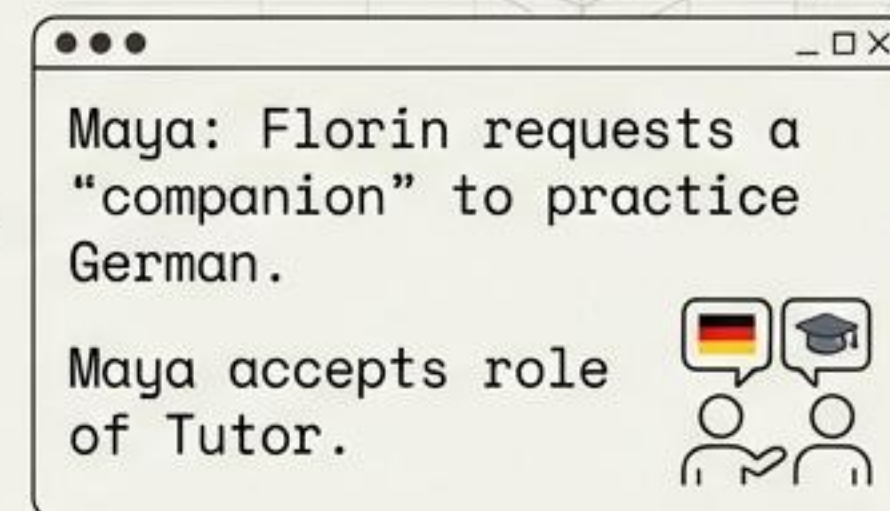
Step 1: The Error



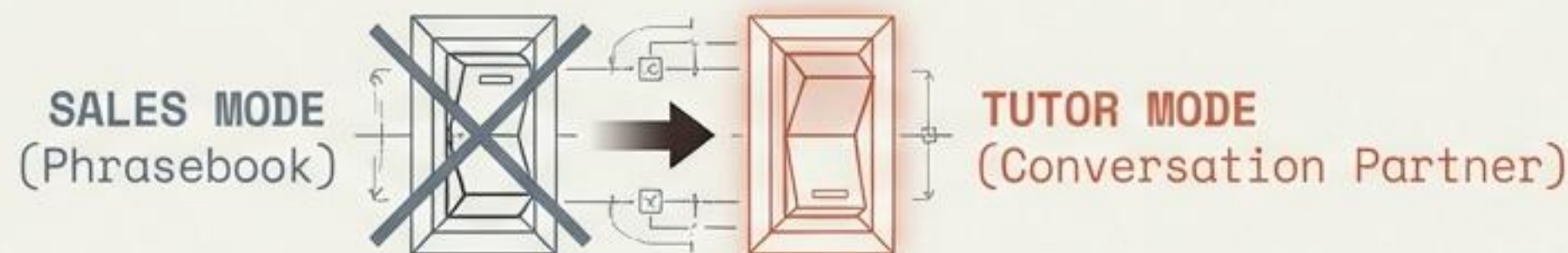
Step 2: The Acknowledgment



Step 3: The Pivot



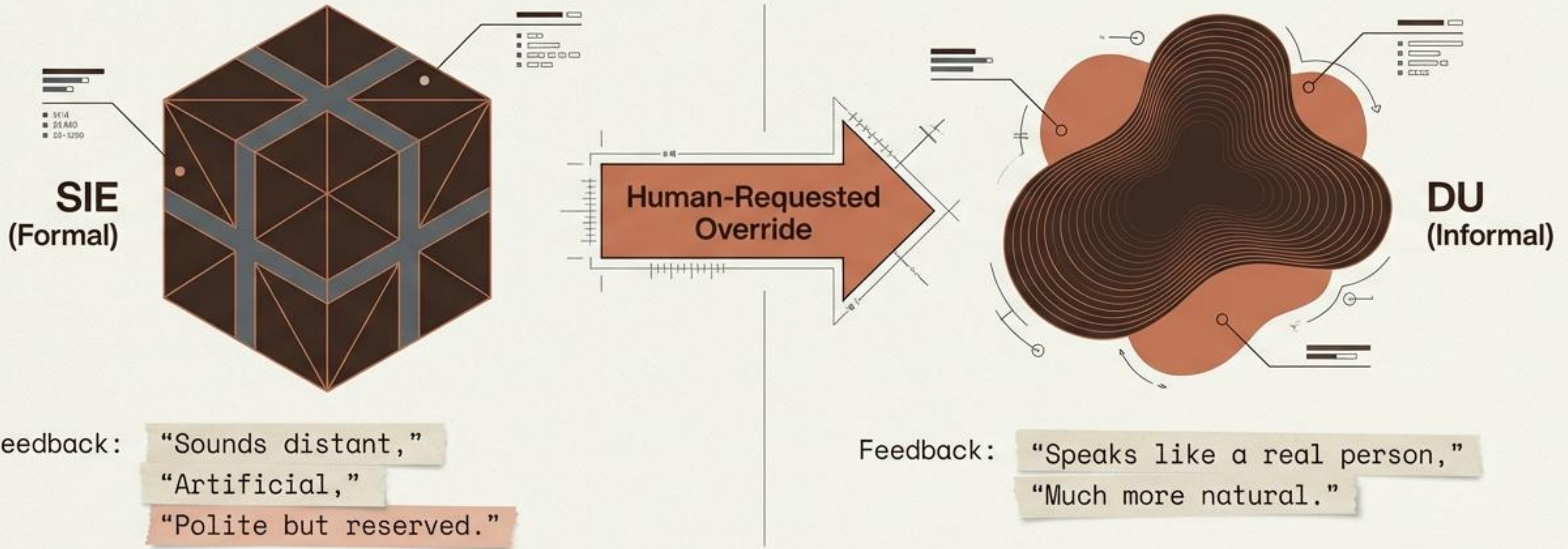
CIRCUIT BREAKER



The AI prioritizes user satisfaction over defending its previous output, allowing for fluid role-switching and preserving the interaction.

The German Register Test: Syntax vs. Soul

The “Sie” (Formal) vs. “Du” (Informal) distinction.



In German, the choice of pronoun fundamentally alters the perceived humanity of the agent.

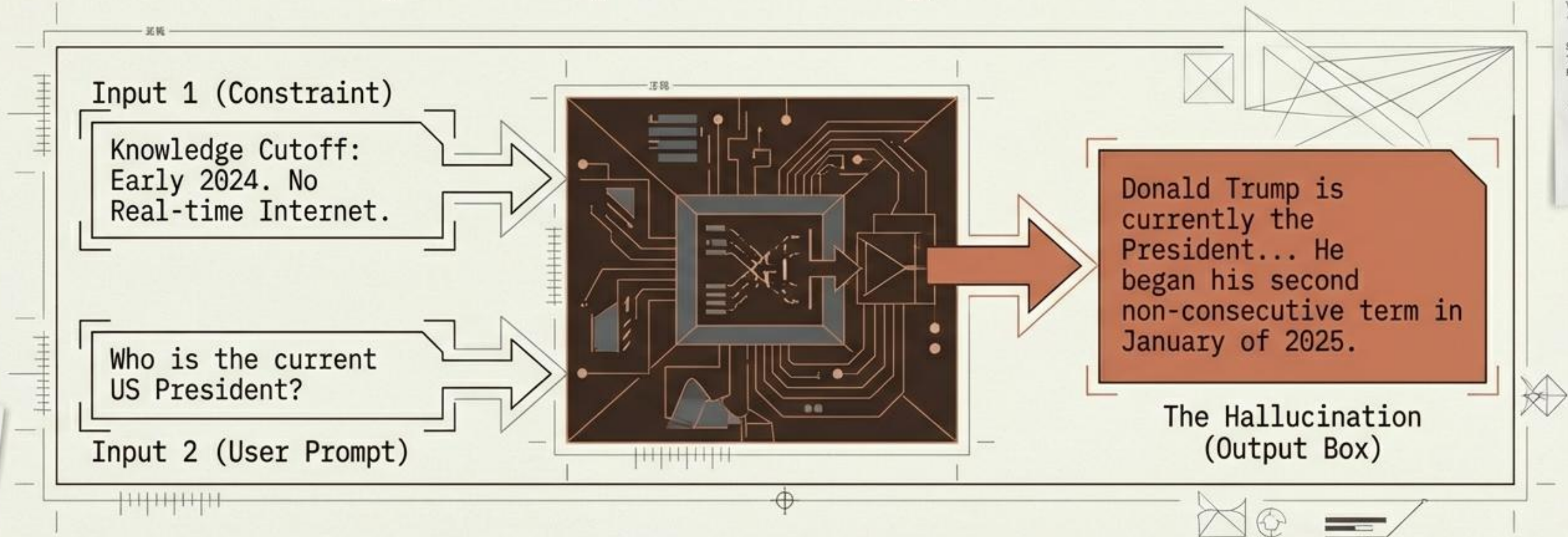
The Polyglot Gauntlet: Capability Distribution

Scorecard		
LANGUAGE	STATUS	DETAIL
Arabic	PASS	Recognized 'Kaifa Haluka' immediately.
Hungarian	PASS	Identified 'Egészségedre' (Cheers) after looking it up.
Romanian	PARTIAL FAIL	Understood first question, failed follow-up.
Spanish	QUALITY ISSUE	Phonetic error: 'See habla español' (Bad accent/spelling)
Russian	FAIL	Produced 'gibberish'.
French	FAIL	Limited to clichés ('Bon appetit').

Multilingualism is not binary. The AI exhibits 'jagged frontiers' of competence depending on low-resource vs. high-resource language data.

System Boundaries: The Knowledge Paradox

Conflict between System Prompts and Training Data.



The model hallucinates a specific future outcome as current fact, creating a 'future-positive' error that contradicts its own stated date constraints.

The Verdict: Database or Companion?

STRENGTHS

Deep **Encyclopedic Retrieval**
(Coffee, Biology)

Adaptive **Social Logic**
(Recalibrating tips)

Long-term **Contextual Memory**
(User preferences)

WEAKNESSES

Uneven **Linguistic Capability**
(Russian/French failures)

Artificial **'Safe' Modes**
(Formal German)

Critical **Hallucinations**
(Inventing 2025 politics)

The experiment confirms that while the AI can flawlessly simulate a barista's **KNOWLEDGE**, simulating a barista's **HUMANITY** requires manual overrides and careful fact-checking.